

Data Communications

Transport Protocols - TCP



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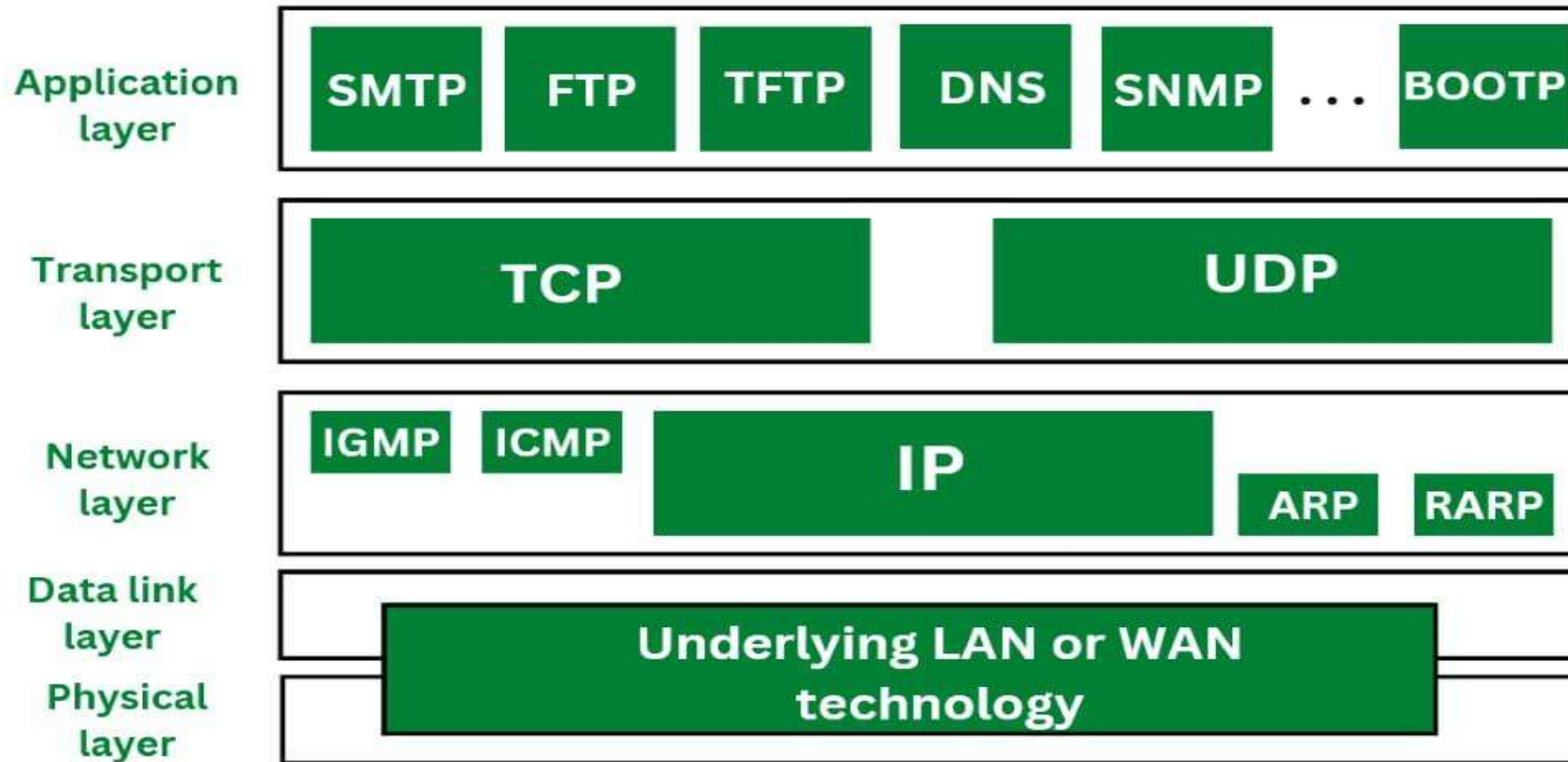
Transport Protocols

Introduction

- The transport layer is the fourth layer in the OSI model and the second layer in TCP/IP model.
- The transport layer provides with end to end connection between the source and the destination and reliable delivery of the services. Therefore transport layer is known as the **end-to-end layer**.
- The transport layer takes the services from its upward layer which is the application layer and provides it to the network layer.
- **Segment** is the unit of data encapsulation at the transport layer.

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Transport Protocols



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Transport Protocols - TCP

- TCP stands for Transmission Control Protocol.
- It provides full transport layer services to applications.
- It is a connection-oriented protocol. i.e., the connection established between both the ends of the transmission.
- For creating the connection, TCP generates a virtual circuit between sender and receiver for the duration of a transmission.

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Transport Protocols – TCP - Features

- **Stream data transfer:** TCP protocol transfers the data in the form of contiguous stream of bytes. TCP group the bytes in the form of TCP segments and then passed it to the IP layer for transmission to the destination.
- **Reliability:** TCP assigns a sequence number to each byte transmitted and expects a positive acknowledgement from the receiving TCP. If ACK is not received within a timeout interval, then the data is retransmitted to the destination. The receiving TCP uses the sequence number to reassemble the segments if they arrive out of order or to eliminate the duplicate segments.

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Transport Protocols – TCP - Features

- **Flow Control:** When receiving TCP sends an acknowledgement back to the sender indicating the number the bytes it can receive without overflowing its internal buffer.
- **Multiplexing:** Multiplexing is a process of accepting the data from different applications and forwarding to the different applications on different computers. At the receiving end, the data is forwarded to the correct application. This process is known as demultiplexing.
- **Logical Connections:** The combination of sockets, sequence numbers, and window sizes, is called a logical connection. Each connection is identified by the pair of sockets used by sending and receiving processes.

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Transport Protocols – TCP - Features

- **Full Duplex:** TCP provides Full Duplex service, i.e., the data flow in both the directions at the same time. To achieve Full Duplex service, each TCP should have sending and receiving buffers so that the segments can flow in both the directions. TCP is a connection-oriented protocol. Suppose the process A wants to send and receive the data from process B. The following steps occur:
 - Establish a connection between two TCPs.
 - Data is exchanged in both the directions.
 - The Connection is terminated.

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Transport Protocols – TCP – Header

Source port address 16 bits					Destination port address 16 bits				
Sequence number 32 bits									
Acknowledgement number 32 bits									
HLEN 4 bits	Reserved 6 bits	URG	ACK	PUSH	RESET	SYN	FIN	Window size 16 bits	
Checksum 16 bits					Urgent pointer 16 bits				
Options & padding									

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Transport Protocols – TCP - Header

- **Source port(16 bits):** It is the address of the application program in a source computer.
- **Destination port(16 bits):** the address of the application program in destination computer.
- **Sequence number(32 bits):** The sequence number field represents the position of the data in an original data stream.
- **Acknowledgement number(32 bits):** contains the sequence number of the next data that the TCP entity expects to receive.
- **Header Length (HLEN):** It specifies the size of the TCP header. The maximum size of the TCP header is 60 bytes, and the minimum size of the TCP header is 20 bytes.
- **Reserved:** It is a six-bit field which is reserved for future use.

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Transport Protocols – TCP - Header

There are total six types of flags in control field:

- **URG:** The URG field indicates that the data in a segment is urgent.
- **ACK:** When ACK field is set, then it validates the acknowledgement number.
- **PSH:** The PSH field is used to inform the sender that higher throughput is needed so if possible, data must be pushed with higher throughput.
- **RST:** The reset bit is used to reset the TCP connection when there is any confusion occurs in the sequence numbers.
- **SYN:** The SYN field is used to synchronize the sequence numbers.
- **FIN:** The FIN field is used to inform the receiving TCP module that the sender has finished sending data.

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Transport Protocols – TCP - Header

- **Window Size:** The window is a 16-bit field that defines the size of the window.
- **Checksum:** The checksum is a 16-bit field used in error detection.
- **Urgent pointer:** If URG flag is set to 1, then this 16-bit field is an offset from the sequence number indicating that it is a last urgent data byte.
- **Options and padding:** It defines the optional fields that convey the additional information to the receiver.



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Transport Protocols - UDP



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Transport Protocols - UDP

- UDP stands for **User Datagram Protocol**.
- UDP is a simple protocol and it provides non sequenced transport functionality.
- UDP is a connectionless protocol.
- UDP is used when reliability and security are less important than speed and size.
- UDP is an end-to-end transport level protocol that adds transport-level addresses, checksum error control, and length information to the data from the upper layer.
- The packet produced by the UDP protocol is known as a **user datagram**.

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Transport Protocols – UDP Header

Source port address 16 bits	Destination port address 16 bits
Total Length 16 bits	Checksum 16 bits
Data	

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Transport Protocols - UDP

- **Source port address:** It defines the address of the application process that has delivered a message. The source port address is of 16 bits address.
- **Destination port address:** It defines the address of the application process that will receive the message. The destination port address is of a 16-bit address.
- **Total length:** It defines the total length of the user datagram in bytes. It is a 16-bit field.
- **Checksum:** The checksum is a 16-bit field which is used in error detection.

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Transport Protocols – TCP & UDP Comparison

Differences b/w TCP & UDP

Basis for Comparison	TCP	UDP
Definition	TCP establishes a virtual circuit before transmitting the data.	UDP transmits the data directly to the destination computer without verifying whether the receiver is ready to receive or not.
Connection Type	It is a Connection-Oriented protocol	It is a Connectionless protocol
Speed	slow	high
Reliability	It is a reliable protocol.	It is an unreliable protocol.
Header size	20 bytes	8 bytes
acknowledgement	It waits for the acknowledgement of data and has the ability to resend the lost packets.	It neither takes the acknowledgement, nor it retransmits the damaged frame.

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Internet applications: Internet Directory Services - DNS



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Internet Directory Services - DNS

The Domain Name Service (DNS) is a directory look up service that provides a mapping between the name of a host on the internet and its numerical address.

- DNS is required for the functioning of the internet.
- DNS is a service that translates the domain name into IP addresses. This allows the users of networks to utilize user-friendly names when looking for other hosts instead of remembering the IP addresses.
- For example, suppose the FTP site at EduSoft had an IP address of 132.147.165.50, people would reach this site by specifying ftp.EduSoft.com. Thus, the domain name is more reliable than IP address.

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Internet Directory Services - DNS

- DNS is a TCP/IP protocol used on different platforms.
- The domain name space is divided into three different sections:
 - generic domains**
 - country domains and**
 - inverse domain.**

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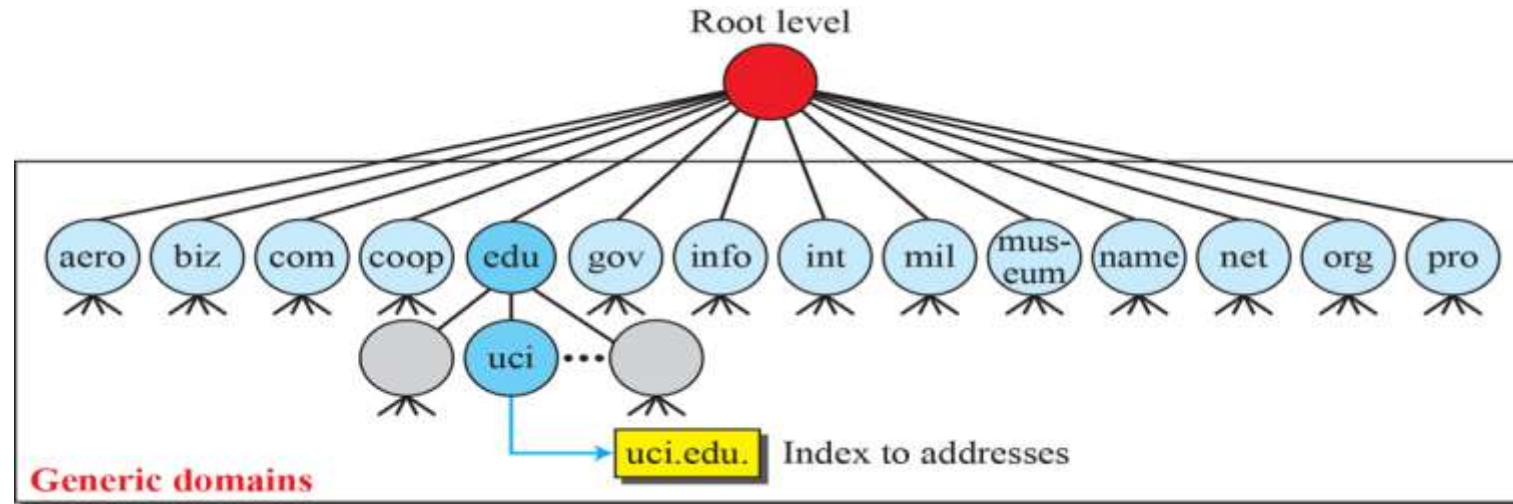
Internet Directory Services - DNS

Generic Domains

- It defines the registered hosts according to their generic behavior.
- Each node in a tree defines the domain name, which is an index to the DNS database.
- It uses **character labels**, and these labels describe the organization type.

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Internet Directory Services - DNS



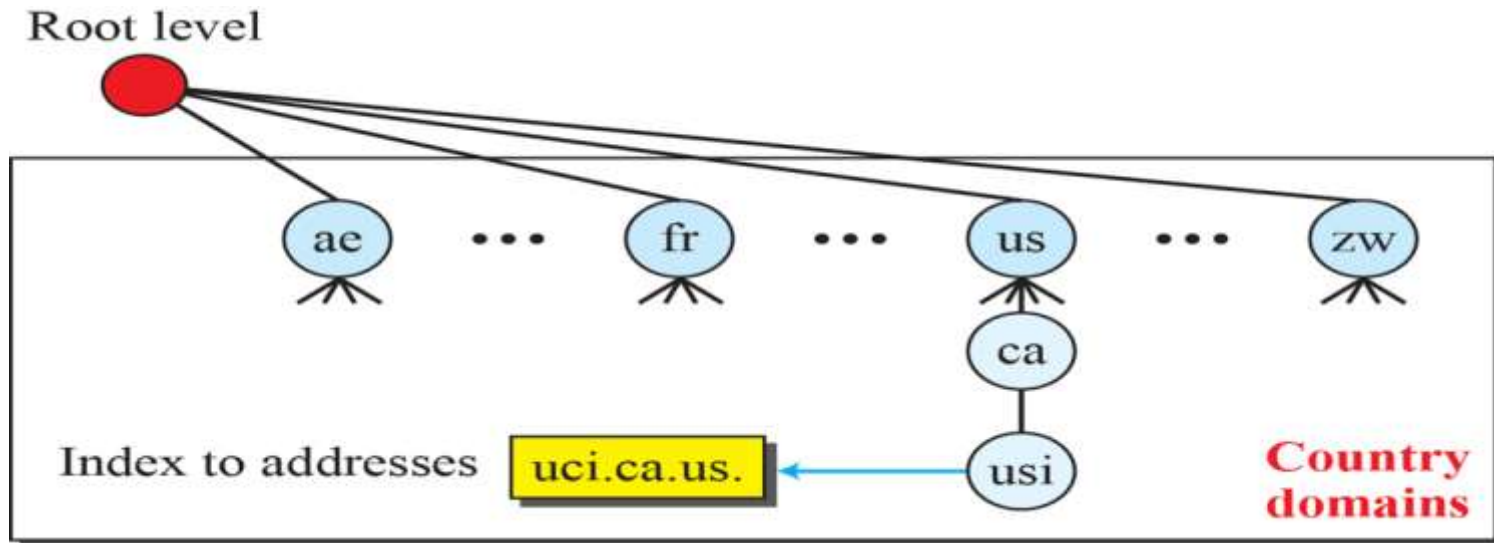
<i>Label</i>	<i>Description</i>	<i>Label</i>	<i>Description</i>
aero	Airlines and aerospace	int	International organizations
biz	Businesses or firms	mil	Military groups
com	Commercial organizations	museum	Museums
coop	Cooperative organizations	name	Personal names (individuals)
edu	Educational institutions	net	Network support centers
gov	Government institutions	org	Nonprofit organizations
info	Information service providers	pro	Professional organizations

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Internet Directory Services - DNS

Country Domains

The format of country domain is same as a generic domain, but it uses two-character country abbreviations (e.g., us for the United States) in place of three character organizational abbreviations.



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Internet Directory Services - DNS

Inverse Domains

- Inverse domain is used to map an address to a name.
- **For example**, a client send a request to the server for performing a particular task, server finds a list of authorized client. The list contains only IP addresses of the client.
- The server sends a query to the DNS server to map an address to a name.
- This query is called an inverse query.

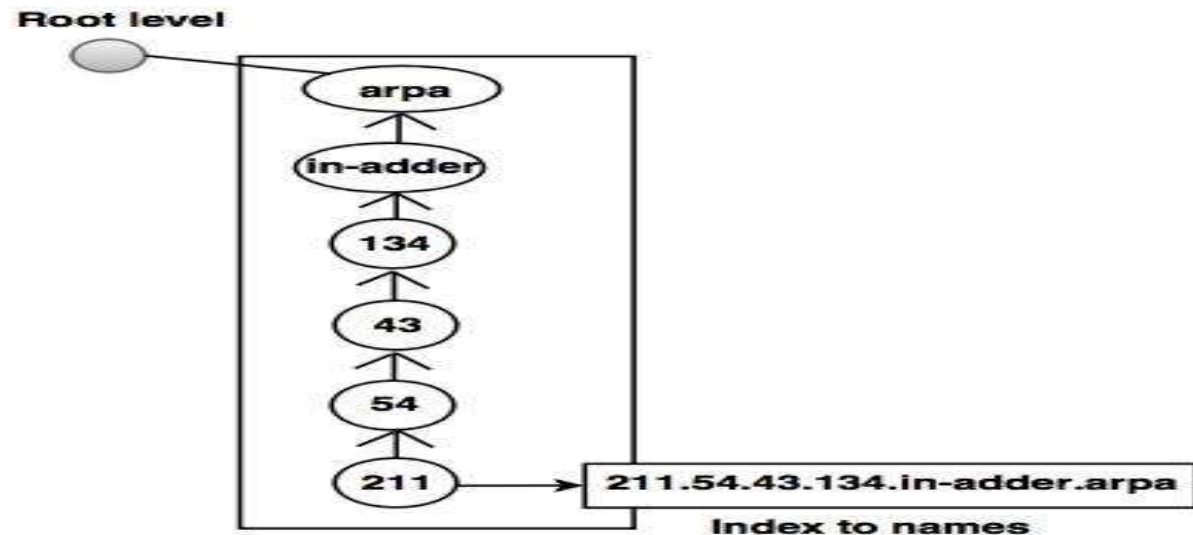
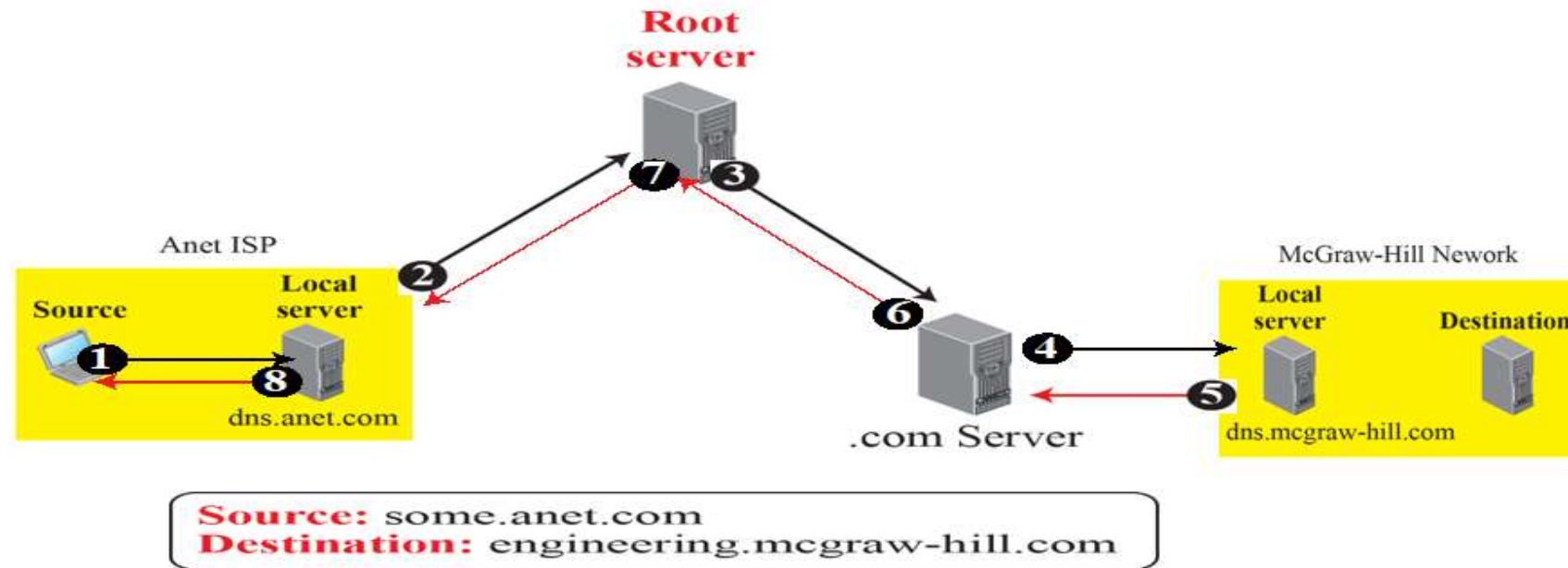


Fig. Inverse domain

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Internet Directory Services – DNS Operation(Recursive)

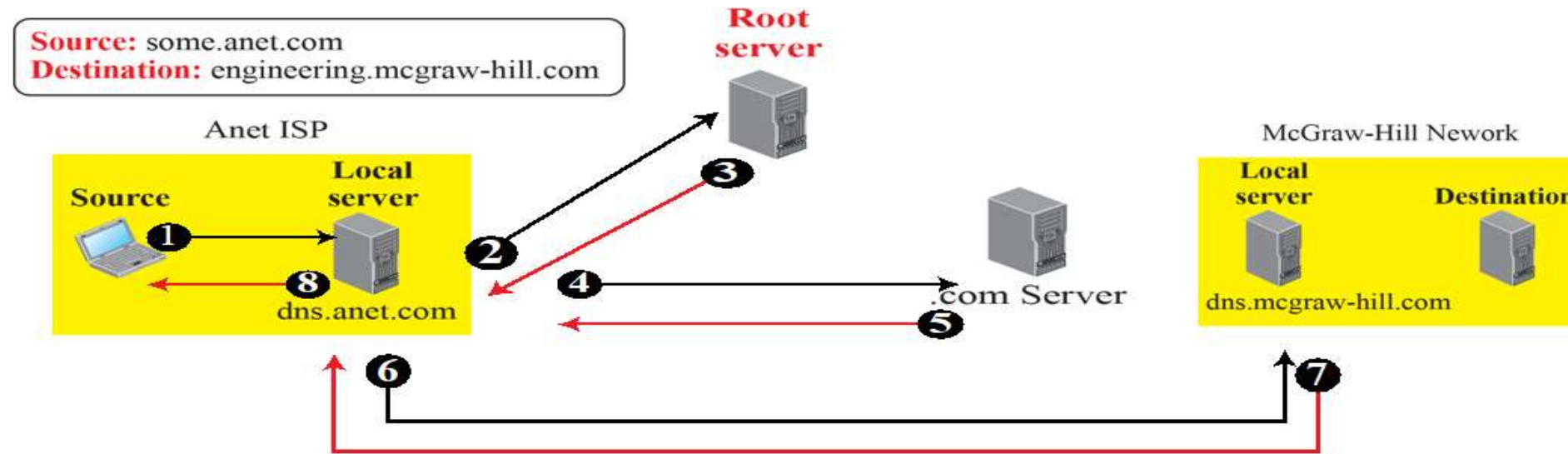
Name – address resolution is done by DNS by **Iterative** or **Recursive** method



Here the query is passed on to the next server in recursive manner

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Internet Directory Services – DNS operation(Iterative)



- Here each server that does not know the mapping sends the IP address of the next server back to the requester. Here events 2, 4, 6 contains the same query. Only source to local DNS is recursive, all other steps are iterative in nature
- In practice, typically iterative pattern is used

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Internet applications: Web Access - HTTP



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HTTP

HTTP (Hypertext Transfer Protocol) is a fundamental protocol of the Internet, enabling the transfer of data between a client and a server.

- It is the foundation of data communication for the World Wide Web.
- HTTP provides a standard between a web browser and a web server to establish communication.
- It is a set of rules for transferring data from one computer to another.
- Data such as text, images, and other multimedia files are shared on the World Wide Web. Whenever a web user opens their web browser, the user indirectly uses HTTP.

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Web Access - HTTP

- It is an application layer protocol that is used for distributed, collaborative, hypermedia information systems.



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HTTP Messages

- HTTP has **Request and Response Message**
- There are four sections for the HTTP Messages

Request line / Status line

Header lines

Blank line

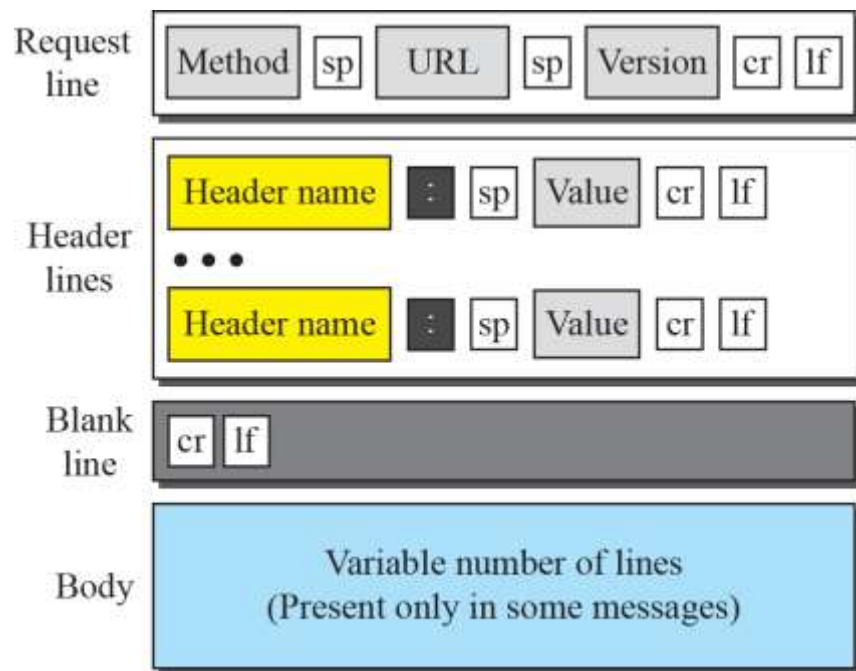
Body

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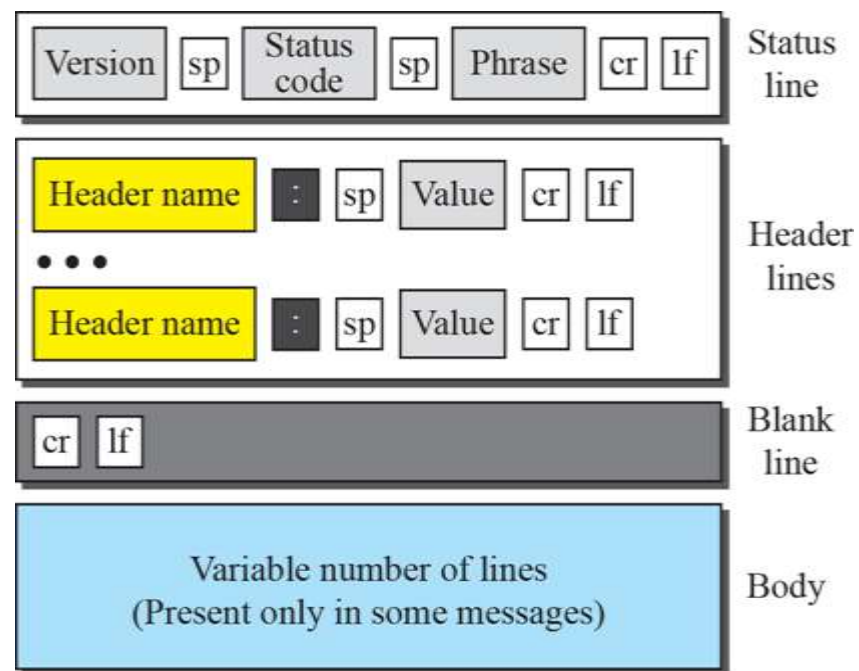
HTTP Messages

HTTP has Request and Response Message

Legend sp: Space cr: Carriage Return lf: Line Feed



Request message



Response message

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HTTP Methods

Methods of HTTP

<i>Method</i>	<i>Action</i>
GET	Requests a document from the server
HEAD	Requests information about a document but not the document itself
PUT	Sends a document from the client to the server
POST	Sends some information from the client to the server
TRACE	Echoes the incoming request
DELETE	Removes the web page
CONNECT	Reserved
OPTIONS	Inquires about available options

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HTTP Response Status codes

Status code appears in 1st line in server-to-client response message

200 OK request succeeded, requested object later in this msg

301 Moved Permanently requested object moved, new location specified later in this msg

400 Bad Request request msg not understood by server

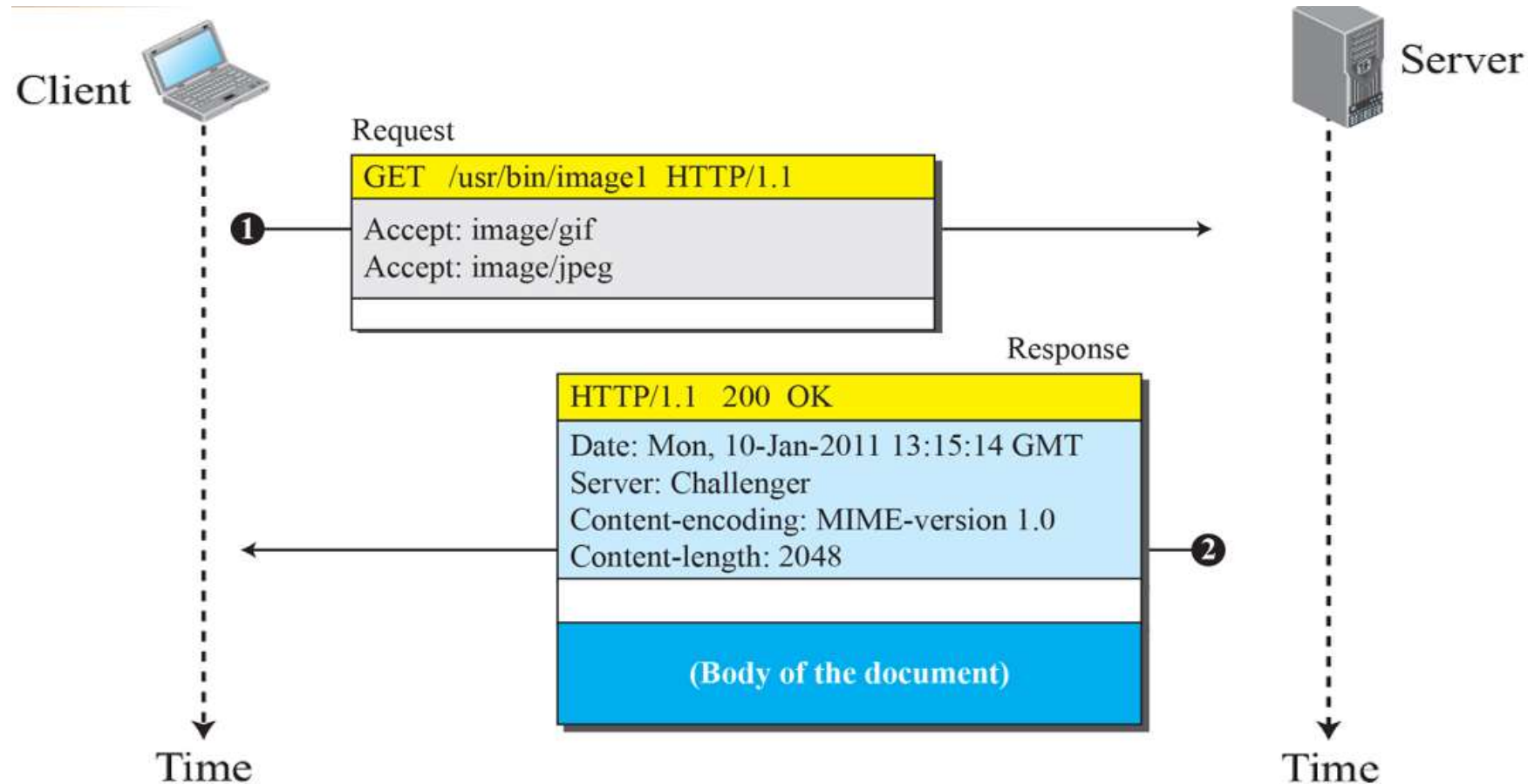
404 Not Found requested document not found on this server

500 Internal Server Error

505 HTTP Version Not Supported

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HTTP Example



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Multimedia Applications: Audio & Video Compression



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Audio & Video Compression



Audio/Video compression is the process of reducing the file size of an audio/video file while attempting to preserve as much of its quality as possible.

There are two main types of compression – ***Lossy and Lossless compression.***

Lossy Compression:

Lossy compression reduces file size by permanently removing audio data that is considered less audible to the human ear. This results in some loss of quality, but it can achieve much higher compression ratios. In case of image or video, the criterion is that the decompressed image is distinguishable from the original to the human eye.

Example: MP3, AAC, Ogg Vorbis, Opus etc

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Audio & Video Compression

Lossless Compression:

- Lossless compression reduces the file size without losing any data,
- The original quality is perfectly preserved.
- It is useful for archiving or when audio/video fidelity is a priority.

Example: FLAC (Free Lossless Audio Codec), ALAC (Apple Lossless Audio Codec), WAV (Waveform Audio File Format), AIFF (Audio Interchange File Format)

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Audio & Video Compression

Advantages and Disadvantages:

- **Lossy Compression:**
 - **Pros:** Drastically reduces file size, suitable for casual listening and streaming.
 - **Cons:** Loss of audio fidelity, noticeable artifacts at lower bit rates.
- **Lossless Compression:**
 - **Pros:** Preserves full audio/video quality, ideal for audiophiles or archival purposes.
 - **Cons:** Larger file sizes compared to lossy formats.

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Audio Compression

MPEG Audio Compression

- It is designed to reduce the size of audio files while preserving sound quality.
- It uses **psychoacoustic models** to remove sounds that are inaudible.

Key Steps in MPEG Audio Compression:

1. Pre-Processing
2. Psychoacoustic Model
3. Quantization
4. Huffman Coding
5. Bitrate Control
6. Multiplexing

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Audio Compression



1. Pre-Processing

Audio input is typically **converted to a frequency domain** (e.g., using **Fourier transform** or **MDCT – Modified Discrete Cosine Transform**) to make it easier to apply compression techniques.

2. Psychoacoustic Model:

This model uses knowledge of human hearing to remove inaudible or redundant parts of the audio signal. For example, sounds masked by louder sounds or outside the range of human hearing are discarded. This is known as **perceptual coding**.

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Audio Compression



3. Quantization:

The frequency-domain coefficients are quantized, meaning they are rounded to fewer bits. This step introduces **loss** but significantly reduces data size.

4. Huffman Coding:

This is an entropy encoding technique where more frequently occurring audio patterns are assigned shorter codes, further reducing the size of the audio data.

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Audio Compression

5. Bitrate Control:

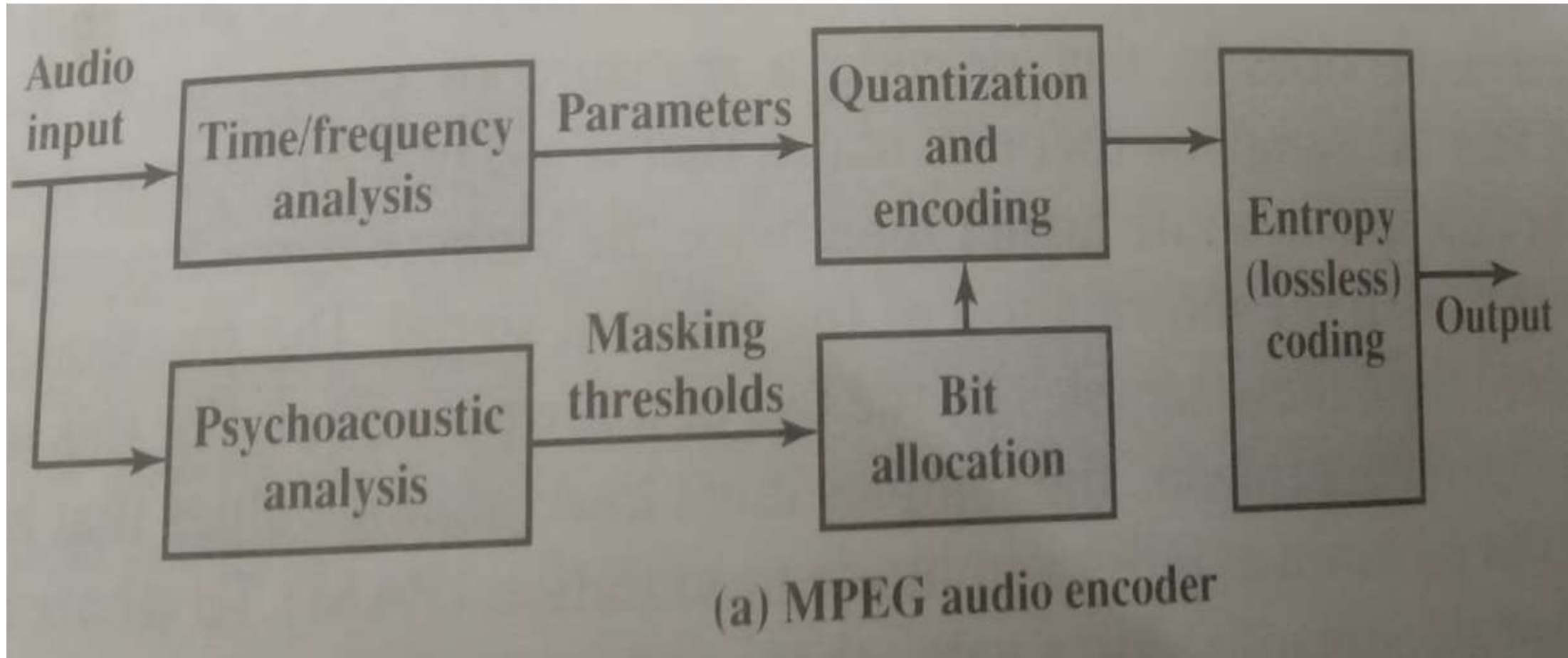
- The encoder adjusts the amount of data used for encoding depending on the desired output quality and target bitrate. Higher bitrates generally result in better quality audio.

6. Multiplexing:

- The compressed audio data is multiplexed with the video stream into a single MPEG container (e.g., MPEG-2 or MPEG-4), allowing synchronized playback of video and audio.

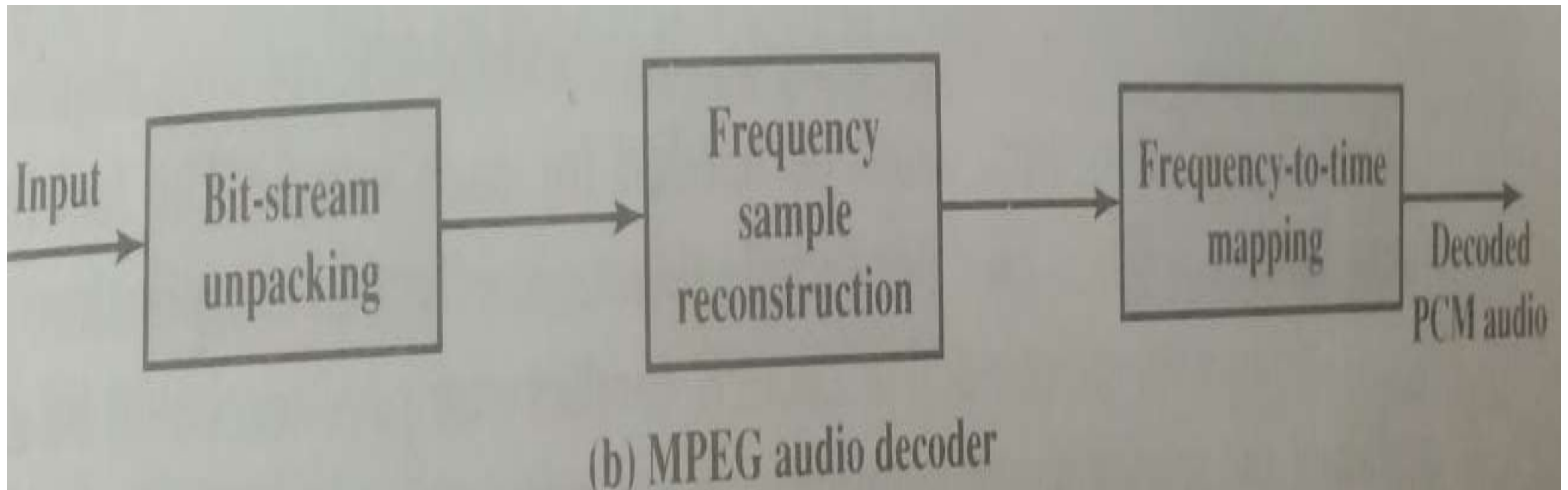
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Audio Compression



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Audio Compression



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Multimedia Applications: Video Compression



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MPEG Video Compression

MPEG video compression is based on two key concepts:

- **Spatial Compression (Intra-frame):** This reduces redundancies within a single frame (image).
- **Temporal Compression (Inter-frame):** This reduces redundancies between consecutive frames.

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Video Compression

Key Steps in MPEG Video Compression:

- 1. Pre-Processing:** The video frame is prepared, including tasks like color space conversion (e.g., from RGB to YCbCr) and downscaling (if needed).
- 2. Motion Estimation:** The encoder analyzes the video and identifies areas where pixel movement occurs between frames. This information is stored as motion vectors, which describe how parts of the frame have moved relative to other frames.

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Video Compression



3. **Motion Compensation:** Using the motion vectors, the encoder predicts parts of the current frame by referring to previous frames, which helps avoid encoding the same information repeatedly.

4. **Intra-frame Coding (Spatial Compression):** For I-frames (Intra-coded frames), the encoder compresses each frame independently, using techniques like Discrete Cosine Transform (DCT) and Quantization. This step reduces spatial redundancies within each individual frame.

5. Inter-frame Coding (Temporal Compression):

P-frames (Predicted frames) and B-frames (Bidirectionally predicted frames) are encoded using the motion vectors calculated earlier. These frames store differences between the predicted frame and the actual frame.

P-frames use information from previous frames to predict the current frame.

B-frames use information from both previous and subsequent frames to make predictions, resulting in even greater compression.

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Video Compression



6. Quantization: The Discrete cosine transformation (DCT) values are quantized into finite values, this step introduces some loss in quality but is essential for reducing file size.

7. Entropy Coding: After quantization, entropy coding techniques such as Huffman coding or Arithmetic coding are applied to represent the remaining data more efficiently by using variable-length codes for more frequent values.

8. Multiplexing: The final step involves multiplexing the encoded video with audio data into a single file or stream (e.g., an MPEG-2 or MPEG-4 file format). This allows for synchronized playback of video and audio.

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Multimedia Applications: Session Initiation Protocol(SIP)



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Session Initiation Protocol(SIP)

- SIP is a collection of rules governing **multimedia exchange** over the internet.
- The application layer control protocol is responsible for establishing, controlling, and terminating multimedia sessions with one or multiple users.
- The multimedia session may include any type of multimedia, such as data, sound, video, picture, etc.
- SIP function in the same way as HTTP does, using a request and response model.

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Session Initiation Protocol(SIP)

- SIP supports **trunking**, which allows to connect to a traditional phone network, like a landline or mobile device.
- It permits to call any phone number, whether it is connected to the internet or not.
- It is more flexible and user-friendly phone system when integrated into a VoIP system.

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Session Initiation Protocol(SIP)

Working of SIP

- Firstly, the SIP caller makes a request and sends an invite message to the receiver.
- After that, the proxy server examines the message body structure, which includes media type, format, and the caller's capabilities.
- If the callee approves the request, the caller receives the response code. The callee may also use the OPTIONS mechanism to inquire about the host's capabilities and other information.

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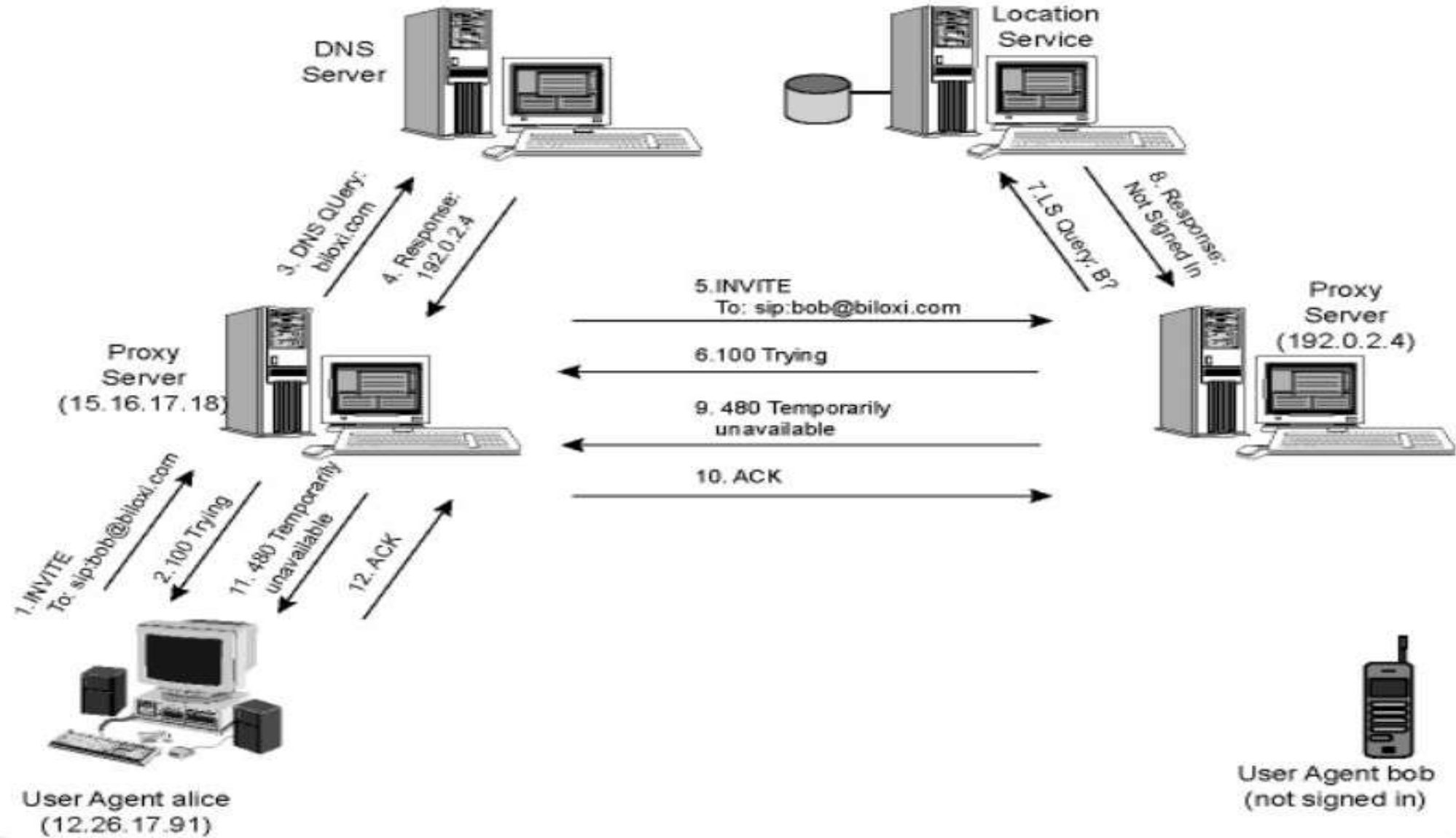
Session Initiation Protocol(SIP)

Working of SIP

- The connection is finished utilizing the three-way handshaking protocol.
- After that, the caller sends an ACK message to end the protocol and confirm receipt of the 200 (OK) messages.
- In the end, the session is ended by sending a BYE method to any party.

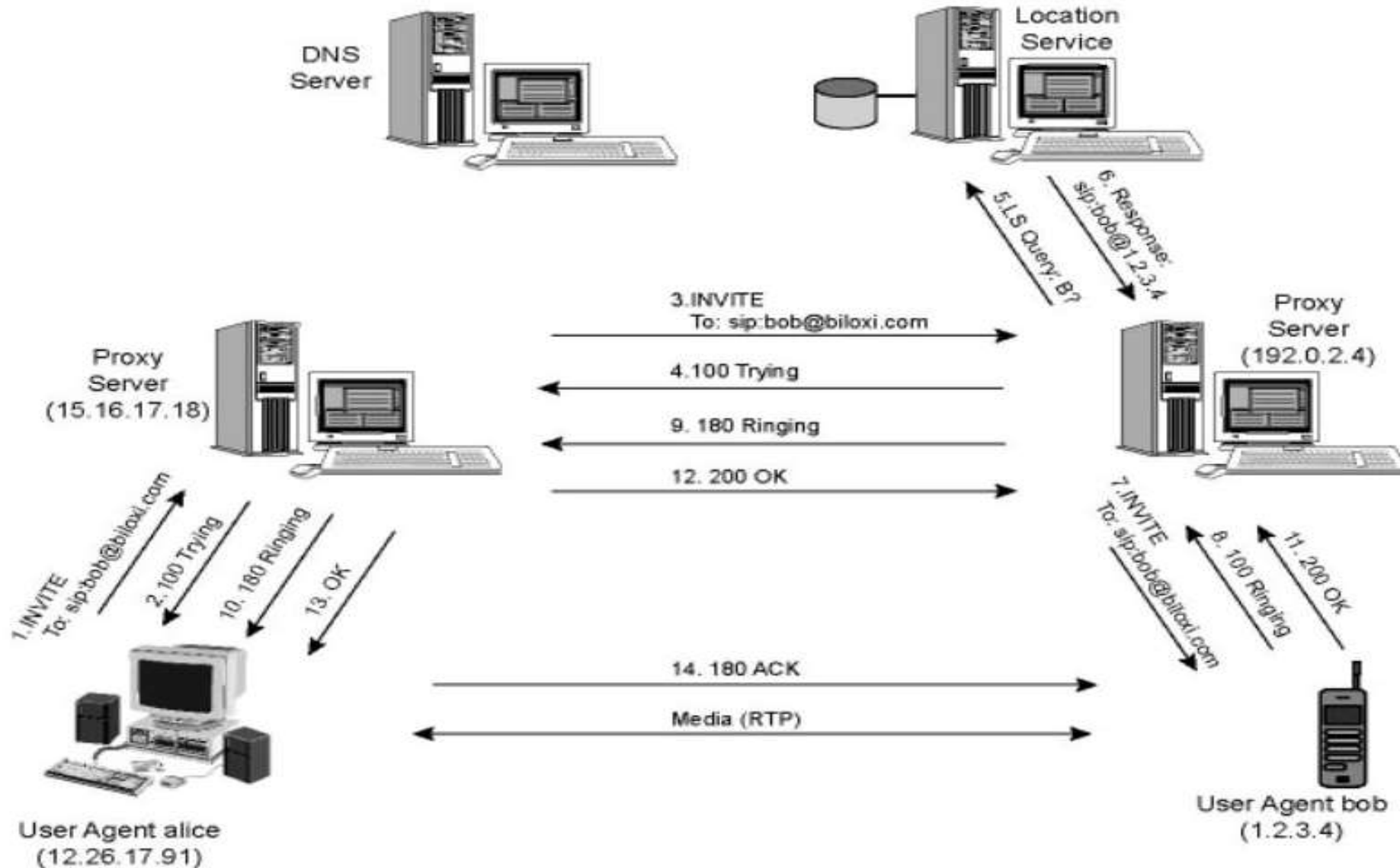
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SIP Call Setup Attempt Scenario



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SIP Successful Call Set Up



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Session Initiation Protocol(SIP)

Component of SIP Protocol - There are four types of SIP components.

1. **User Agents** - The client and server are both user agents, where the client makes the requests, and the server receives and responds to them.
2. **Several Servers** - The SIP protocol uses various servers, including proxy, location, registrar, and redirect servers. Each server works on its own criteria.
3. **Gateways** - A gateway is just a user agent that can connect to other networks.
4. **B2B(Business-to-Business) User Agents** - It includes two user agents that may transmit and alter SIP messages.

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Session Initiation Protocol(SIP)

Advantages:

- It may support numerous users with different skills.
- For example, in a conference with some users who have video capability and others who only have audio capabilities, the video users may see each other. They are not required to use exclusively audio, as with other protocols.
- It operates independently of the session type or medium used and provides flexibility.

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Session Initiation Protocol(SIP)

Disadvantages

- Fewer people are familiar with this protocol because it is a relatively new protocol. To use SIP protocol within the network, make sure that we have trained support staff.
- SIP features are continually being developed, although many companies have their own proprietary protocol implementations.

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Multimedia Applications: Voice over Internet Protocol(VoIP)



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Voice over IP (VoIP)

- VoIP technology enables voice calls to be made utilizing a broadband Internet connection rather than a traditional phone line.
- VoIP service or device enables you to call many persons simultaneously, while others could enable you to call anyone.
- The call covers long-distance, mobile, local, and international lines.
- Some VoIP services insist to utilize a particular VoIP phone or computer, others allow to utilize a standard phone connected to a VoIP adaptor.

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Voice over IP (VoIP)

Working of VoIP

- VoIP services turn the voice into a digital signal for Internet transmission.
- When you call a regular telephone number, the signal is changed into a typical telephone signal before reaching the intended destination.
- It allows to place calls from a system, a dedicated VoIP-specific phone, or a regular phone that is linked to a VoIP adaptor.
- The wireless "hot spots" in public spaces like parks, airports, and cafes let you access the internet and may use VoIP services wirelessly.

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Voice over IP (VoIP)

Advantages of VoIP

1. The connection can be offered more smoothly than with an analog signal.
2. It is possible to avoid paying for both an internet connection and a landline.

Disadvantages:

1. Not all VoIP devices utilize emergency service contact numbers to connect to emergency services.
2. Some VoIP devices cannot work properly during power outages
3. VoIP service providers can or cannot offer directory assistance.



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